

TillzoPOS QA FINAL REPORT v2

2026-08-22 | Full Audit Round 2 (Delegation Findings + Reinstall Persistence)

EXECUTIVE SUMMARY

Result:

REINSTALL PERSISTENCE TEST PASS (100%) - uninstall/reinstall/same Gmail login -> 8/8 sales, inventory, vendors, expenses, customers ALL restored from cloud.

Fixes this round:

24 defects fixed + verified at code level (DEF-30,32,33,38,39,40,41,42,43,44,44b,45,46,47,48,49,50,51,52,53,57,60,74,80) + GAP-1 Printer Settings route wired.

Registry:

DEF-01..80 + GAP-1/2 (98 entries) - 7 HIGH + 27 MED + 9 LOW from subagent deep-scan merged, re-numbered.

Live proof:

Sale 14 INVOICE 023A1FCB \$220 CASH -> sheet row 11 (9 sales on sheet). Sync LIVE: 'Remote updates detected', Sales_Aug_2026!1:1 reads. Sales_1:1 HTTP 400 eliminated (DEF-80).

FIXED THIS ROUND (24) - CODE-LEVEL, VERIFIED

- DEF-30 [SyncWorker single-flight - duplicate uploads impossible ('Another SyncWorker is already running - skipping' verified)]
- DEF-32 [Sales header repair - SchemaGuard dynamic Sales_[MMM_YYYY] tabs + blank sync_uuid skip + deleteCorruptSales (History 8/8 clean)]
- DEF-33 [Settings last_updated_timestamp duplicate + loop-0 skip -> delta pull was DEAD; poll now LIVE (remote=local verified)]
- DEF-38 [SchemaGuard 429/empty-response repair guard (no false header repair on rate-limit)]
- DEF-39 [Cash over-tender change now netted from till (Z-report Expected Cash accurate)]
- DEF-40 [GRN confirm idempotent - double-tap no longer double-increments stock]
- DEF-41 [hasBatches set on GRN/adjustment batches (phantom stock inflation eliminated)]
- DEF-42 [Discount clamp [0, subtotal+tax] - negative/over-discount impossible (clamped -50->0, 300->220 verified)]
- DEF-43 [Batch stock updates now PUT to sheet (was append-only -> updates silently dropped)]
- DEF-44 [PO receivedQty increment + status flip SENT->Received on GRN (UI verified)]
- DEF-44b [PO/GRN header updates via updateRowByUuid (existing rows now sync)]
- DEF-45 [Echo-clobber protection on 13 delta tables (local PENDING rows never overwritten by stale sheet copy)]
- DEF-46 [Refund now creates Khata JAMA credit for customer-linked sales]
- DEF-47 [Loyalty points/spend now mark sync_status=pending (reaches sheet)]
- DEF-48 [Till session queries terminal-scoped (multi-terminal day-close reconciles correct session)]
- DEF-49 [openTill idempotent guard (no duplicate OPEN sessions)]
- DEF-50 [Z-report no longer silently skips with no open session - explicit error]
- DEF-51 [NET IN DRAWER includes opening cash (was sales-expenses only)]
- DEF-52 [Expense update/delete re-adjusts till drawer (delta applied)]
- DEF-53 [createNewSheet includes Time_Clock tab (was missing -> punches never synced on new workspace)]
- DEF-57 [Time Clock IN/OUT state machine fixed (was stuck on IN after OUT)]
- DEF-60 [Time_Clock tab in workspace provisioning]
- DEF-74 [DbEncryption fallback migration-safe (derivable key fixed; existing DBs keep legacy key to avoid 'file is not a database' crash - observed & fixed)]
- DEF-80 [SchemaGuard hardcoded 'Sales_1:1' -> HTTP 400 every check; now resolves real Sales_Aug_2026 from metadata (log verified)]
- GAP-1 [PrinterSettingsScreen was ORPHANED (no nav route - printing unconfigurable). Now Settings > App Info > Printer Settings:

Bluetooth MAC + Wi-Fi IP + test buttons (UI verified)]

GAP-2 (OPEN) - BarcodeScannerScreen unwired

- Full-screen barcode scanner screen exists but is not routed from the POS home scanner. MEDIUM - wiring pending.

REINSTALL PERSISTENCE TEST - PASS

- adb uninstall com.tillzo.pos -> install -> camera Allow -> Continue with Google -> Accept & Sync
- Restored: 8/8 sales (A0B82B17 \$220, 47713D33 \$220, 845ED833 \$100, 136CFE36 \$200, 60EF9E52 \$220, 775425E4 \$500, C94ACB74 \$490, 0D0AD550 \$110)
- Restored: Inventory 2 (PROD-001 stock 12.0 - GRN +10 applied), Vendors 2, Expenses 10, Customers 1
- Data Viewer counts confirm: Sales 8, Inventory 2, Customers 1, Expenses 10

KEY OPEN DEFECTS (TRIAGED, NEXT ROUND)

- DEF-07: PIN unlimited attempts, no lockout (brute-forceable)
- DEF-08: No barcode checksum validation (EAN/UPC)
- DEF-31: Vendors sheet duplicate row 3 (app-route cleanup pending)
- DEF-79: Keystore getEncoded null on hardware-backed keys (Android 10 emulator) - fallback works, data safe
- DEF-50s: Scattered old sales in sheet cols 20-100 (col A empty - skip logic ignores them; cleanup pending)

METHOD

- Every CRUD op: ADD -> Force Sync -> xlsx re-download -> openpyxl row-level verify -> UPDATE -> sync -> verify -> DELETE -> sync -> verify
- Canonical verify route: Desktop Chrome (Forex Studio) xlsx export; emulator Chrome disabled (pm disable-user) to stop download-dialog interference
- Subagent deep-scan (51 api_calls, 20,585 chars) DEF-30..69 + feature-gap audit merged into registry after re-numbering (manual DEF-30..34 collision resolved)
- 30 files patched, +744/-96 lines, no git commit/push (per protocol)